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ORIGINAL ARTICLE

Bidirectional feature fusion via cross-attention transformer for chrysanthemum classification

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Abstract

Chrysanthemums hold significant ornamental, economic, and medicinal value, with their quality and economic worth heavily influenced by geographic origin. Accurate classification of chrysanthemums is crucial for ensuring product authenticity, boosting consumer trust, and promoting sustainable industry growth. Traditional classification methods, however, suffer from inefficiency and high costs. To address these challenges, we propose a novel chrysanthemum classification method utilizing a bidirectional feature fusion approach via cross-attention and two-stream network fusion. Our method preprocesses front and back images of chrysanthemums from diverse regions, employing the powerful Swin Transformer as the backbone to extract features. The cross-attention mechanism effectively integrates features from both image sides, and a secondary training strategy further enhances the model's generalization capabilities. Experimental results demonstrate that our method achieves higher accuracy, precision, recall, and F1 score compared to state-of-the-art models, highlighting its potential for accurate chrysanthemum origin tracing. The code and datasets are openly available at https://github.com/dart-into/CCMCAM, ensuring transparency and reproducibility of our findings.

Keywords Chrysanthemums classification · Swin transformer · Two-stream network · Cross attention · Deep learning

1 Introduction

Chrysanthemums have unique medicinal and ornamental value. Chrysanthemums contain a variety of chemical compounds, primarily flavonoids [1], volatile oils [2], phenylpropanoids, terpenes [3], caffeoylquinic acids, and amino acids. Among these, flavonoids and phenylpropanoids serve as the main active components. Modern pharmacological research indicates that chrysanthemums exhibit a range of pharmacological activities, including the treatment of coronary heart disease, blood pressure reduction. They are also effective in preventing hyperlipidemia and possess

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antibacterial, antioxidant [4], anti-inflammatory [5], and anti-aging effects. Due to differences in climate, soil, and cultivation practices, chrysanthemums from various regions may show significant variations in quality and medicinal efficacy. Therefore, accurate identification of chrysanthemum species and origin not only aids in recognizing high-quality varieties but also ensures product quality. This, in turn, boosts consumer confidence and supports sustainable industry growth.

Traditional methods for determining the origin of chrysanthemums primarily include plant phenotypic analysis and physicochemical analysis. Plant phenotypic analysis involves observing the external features of chrysanthemums, along with sensory factors such as smell and taste, to determine their origin. However, this method is subjective, reliant on experienced experts, and inefficient for large-scale classification tasks. Moreover, the high similarity among many chrysanthemum varieties further complicates manual classification efforts. Physicochemical analysis methods [6, 7] mainly determine the geographic origin of chrysanthemums by measuring the content of active compounds, which ensure high sensitivity and strong reproducibility. Although this method provides precise chemical data, its relatively high costs and lengthy sample preprocessing and analysis times may render it unsuitable for rapid and efficient identification.

While traditional chrysanthemum classification methods have partially addressed identification challenges, their limitations have sparked the need for more precise, objective, and efficient classification approaches. New machine learning-based methods have emerged, enabling the recognition and quantification of chrysanthemum morphological traits, such as petal shape and color distribution. In earlier studies, researchers employed image feature extraction techniques such as gray-level co-occurrence matrix [8], hyperspectral imaging [9], and local binary patterns [10] to identify and detect chrysanthemum varieties, yielding promising results. Deep learning, a subfield of machine learning, has particularly excelled in image classification through models like CNNs and the VIT series. These technologies have played a crucial role in plant morphology studies and have been widely applied in fruit pest detection [11], plant disease identification [12], and maturity assessment [13]. Furthermore, these techniques have shown significant success in traditional Chinese medicine detection, including grading and fraud detection in saffron [14], grading of angelica [15], and origin tracing [16]. Similarly, chrysanthemum origin tracing can be realized through these advanced technological methods, warranting further exploration and implementation.

To address these challenges, this paper integrates a cross-attention mechanism with residual modules, designing and constructing a two-stream Swin Transformer network to comprehensively extract and fuse features from both the front and back images of chrysanthemums. Recent studies, including BaGFN [17] and EGDNet [18], have demonstrated the potential of attention mechanisms for feature extraction and balancing. These methodologies inspire the proposed use of a bidirectional feature fusion via cross-attention (hereinafter referred to as BFCA) module in our two-stream network, facilitating robust feature interaction across image streams. Additionally, we propose a secondary training strategy to enhance the network's generalization capabilities and improve its ability to distinguish between similar chrysanthemum categories.

The primary contributions of this paper are as follows:

(1) A two-stream Swin Transformer network is proposed to comprehensively extract features from both the front and back images of chrysanthemums;
(2) A bidirectional feature fusion approach via cross-attention is introduced to enable the interaction and fusion of features from different subnetworks;
(3) A secondary training strategy is designed to effectively enhance the network's ability to distinguish between similar chrysanthemum categories and improve generalization performance.

The remainder of the paper is organized as follows: Section 2 introduces related work on image classification and chrysanthemum species identification. Section 3 provides a detailed description of the proposed network's structure and specifics. Section 4 presents the results of the performance evaluation experiments. Finally, Section 5 provides the conclusion of the paper.

2 Related work

2.1 Image classification methods based on deep network

Image classification is a fundamental challenge in computer vision, with the goal of categorizing images into distinct classes. Over the decades, various traditional machine learning methods have been employed for image classification. However, the advent of deep learning has positioned deep neural networks as one of the most advanced methods for image classification. AlexNet [19], introduced by Krizhevsky et al., dramatically enhanced image classification performance and training speed by incorporating the ReLU activation function and leveraging GPU acceleration. VGGNet [20], proposed by Simonyan and Zisserman, streamlined model design and enhanced classification accuracy by using smaller convolutional kernels and increasing

network depth. GoogLeNet (Inception) [21], introduced by Szegedy et al. employed a modular design combining various convolutional kernels to capture features across different scales, thereby enhancing model expressiveness while maintaining computational efficiency. ResNet (Residual Network) [22], developed by He et al. improved accuracy by significantly increasing network depth, using residual blocks with skip connections to address the vanishing gradient issue in deep neural networks. DenseNet [23] enhanced training efficiency and accuracy by directly connecting each layer to all preceding layers, thereby promoting feature reuse and gradient flow. RegNet [24], a novel convolutional neural network developed by researchers at Stanford University and Facebook AI Research, achieved higher accuracy with fewer parameters by automatically searching for optimal network architectures. CSPNet [25] effectively reduced redundant gradient information and enhanced computational efficiency and model accuracy by dividing feature maps into two parts and cross-fusing them at various network stages. Google's EfficientNet [26] improved model performance through balanced scaling in depth, width, and resolution. It achieved efficient network design by employing a compound scaling method that simultaneously adjusted network depth, width, and resolution. ConvNeXtV2 [27] is an extension of the ConvNeXt [28] architecture, drawing inspiration from MAE [29]. This framework introduces a novel Global Response Normalization (GRN) layer to foster more competitive feature expression among channels in the original ConvNeXt modules, thereby better capturing distinctive channel features.

In the field of chrysanthemum research, deep learning methods are gradually gaining prominence. Yuan et al. [30] proposed a chrysanthemum image phenotypic classification framework based on transfer learning and bilinear convolutional neural networks, using a symmetric VGG16 network as the feature extractor, with global features fed into the classification layer for final categorization. They later adopted an active learning strategy [31] to select and label highly informative, unannotated chrysanthemum samples, combining ResNet50 with CBAM modules to train the classification model, and achieved high-accuracy classification with a minimal number of labeled samples. Liu et al. [32] used an image acquisition device to photograph 103 large chrysanthemum varieties, obtaining a total of 14,000 images. Leveraging transfer learning principles, they developed image recognition models for large chrysanthemum varieties based on VGG16, GoogLeNet, and ResNet50 deep convolutional neural networks (DCNN) and validated the recognition results, though the clustering and visualization of extracted deep features did not reveal clear distribution patterns. Subsequently, they combined AP clustering [33] with deep feature analysis and proposed a multi-information model based on deep learning for identifying and classifying large-flowered chrysanthemums. Huang et al. [34] proposed an improved multi-scale, multi-parallel convolutional neural network named VGG-Inception, based on the traditional VGG16 convolutional neural network. By using a parallel network structure and global pooling layer, they maintained model depth, increased network width, and reduced the number of parameters to just 9% of the original VGG16, while leveraging auxiliary classifiers to prevent gradient vanishing and improve model generalization. Zhang et al. [35] proposed a metric learning-based method that significantly improved the accuracy and feature interpretability of flower variety recognition with limited sample data by incorporating center loss and L2-softmax loss. Deep learning has significantly advanced image classification. In chrysanthemum research, these methods have been tailored for high-accuracy variety recognition.

2.2 Application of ViT in image processing

With the introduction of the attention mechanism, many researchers have applied it to image classification. Hu et al. [36] proposed SENet (Squeeze-and-Excitation Network), which dynamically adjusts the weights of feature maps through an attention mechanism to improve image classification performance. Woo et al. [37] introduced CBAM (Convolutional Block Attention Module), which combines channel attention and spatial attention to optimize the model's focus on different parts of an image, thereby enhancing classification accuracy. In recent years, the success of Transformer [38] architectures in natural language processing, driven by the self-attention mechanism, has spurred their application in image classification. Dosovitskiy et al. [39] introduced the Vision Transformer (ViT), which divides images into fixed-size patches, linearly embeds these patches into a Transformer, and leverages global self-attention to capture image features. Building on ViT, various derivative models have emerged, such as DeiT (Data-efficient Image Transformers) [40], which improves data efficiency and model performance by introducing teacher models and distillation techniques. Microsoft's researchers proposed TinyViT [41], a novel compact ViT that transfers knowledge from large pre-trained models to smaller models through rapid pre-training distillation, allowing smaller models to benefit from extensive pre-training data. Liu et al. [42] developed Swin Transformer (Hereinafter referred to as SwinT), which achieves efficient performance and lower computational complexity in image classification by combining local and global self-attention through a hierarchical shift-window mechanism. Apple's open-source FastViT [43] uses structural re-parameterization and eliminates skip connections to reduce memory access costs,

further applying over-parameterization during training and large kernel convolutions to enhance accuracy, achieving state-of-the-art latency-accuracy trade-offs. Liu et al. [44] propose EfficientViT, a family of high-speed vision transformers that enhance memory efficiency and reduce computational redundancy, achieving high accuracy. Wang et al. [45] re-examined the efficient design of lightweight CNNs and highlighted their potential on mobile devices. By integrating the efficient architecture design of lightweight ViTs, they progressively enhanced standard lightweight CNNs, resulting in RepViT, which demonstrates excellent performance-latency balance.Other works, such as EAPT [46], have enhanced attention mechanisms in transformers to better preserve semantic information in vision tasks, providing valuable techniques for hierarchical attention extraction.

The rise of Transformer-based architectures, such as ViT and its derivatives, has further revolutionized the field of image processing, offering efficient and accurate classification through innovative approaches and model designs.

These studies demonstrate the enormous potential and feasibility of deep learning in chrysanthemum recognition. Current methods primarily focus on the image information of the front side of chrysanthemums. They often neglect the rich features of the back side, which could be crucial for more comprehensive and accurate recognition and classification. In fact, different varieties of chrysanthemums may have similar front-side features, but their back-side features often exhibit significant differences. The full utilization of the image information from both the front and back sides of chrysanthemums can significantly improve the accuracy and reliability of recognition and classification. Moreover, existing methods have not fully leveraged the capabilities of ViT. Therefore, this paper proposes a two-stream neural network that adopts SwinT as the backbone, utilizing its powerful feature representation ability to extract features from both the front and back sides of chrysanthemum images. Through the BFCA module, the two streams of features are fully interacted and fused, enabling more accurate chrysanthemum species identification.

3 Methods

To achieve efficient chrysanthemum species recognition, the specific workflow of this study is illustrated in Fig. 1. It primarily consists of three components: the front-side feature extraction module, the back-side feature extraction module, and the feature fusion and prediction module. Several cross-attention modules are interspersed between these stages to facilitate feature interaction.

3.1 Data preprocessing

Before inputting the chrysanthemum images into the network, they undergo the following preprocessing steps: First, all images are resized to a uniform dimension of 224×224 pixels to mitigate accuracy degradation due to varying image scales. Next, images are randomly rotated left or right by 15° to enhance the diversity of the training set and reduce the risk of overfitting. Subsequently, the mixup data augmentation technique is applied, which generates new training samples by linearly combining two images and their corresponding labels at a certain ratio. This method helps improve the model's robustness and reduces sensitivity to noise. Finally, all samples in the dataset are normalized to accelerate model convergence and enhance generalization.

Mixup creates new training samples by linearly interpolating between input samples and labels during training, as described by the following formulas:

$$\widetilde{x} = \lambda x_i + (1 - \lambda)\, x_j \tag{1}$$

$$\widetilde{y} = \lambda y_i + (1 - \lambda)\, y_j \tag{2}$$

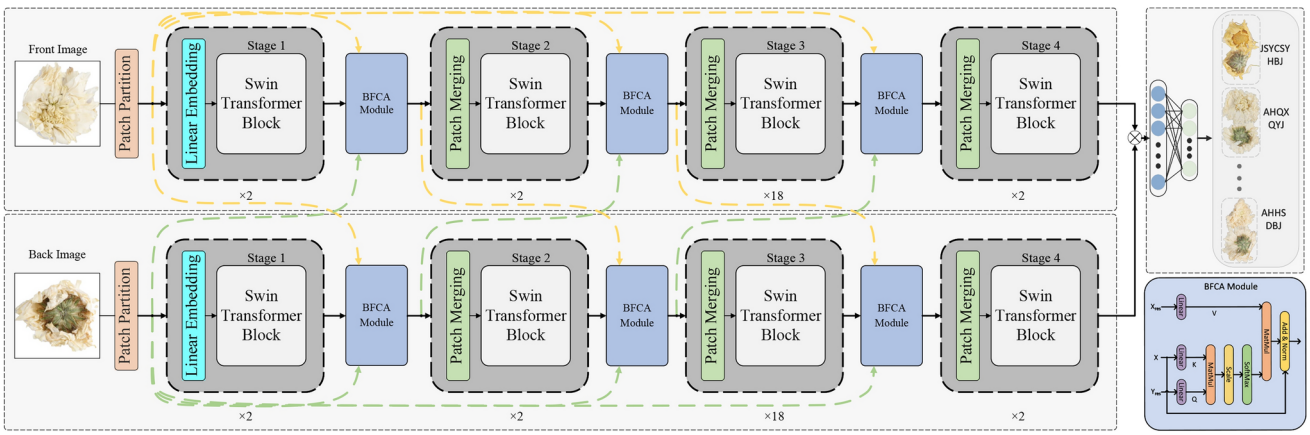


Fig. 1 Network architecture diagram

where x_i and x_j are the input samples, y_i and y_j are the corresponding labels, and the mixing parameter λ is a random number uniformly sampled from the interval [0, 1]. This approach significantly enhances the model's generalization ability and robustness to noisy data by introducing more variation into the sample space.

3.2 Feature extraction

Considering the variability in front and back features among different chrysanthemum species, the proposed method utilizes a two-stream neural network model. Both streams use the SwinT as the backbone, leveraging its powerful feature representation capabilities to extract features from the front and back images of chrysanthemums. The input consists of the front and back images of chrysanthemums, with each stream focusing on extracting critical features specific to the front and back sides. The two-stream design accounts for scenarios where front features might be similar across different species, but back features differ, or vice versa. This approach fully utilizes and integrates the front and back features to achieve more accurate species identification.

The SwinT is composed of four stages, each consisting of 2, 2, 18, and 2 SwinT blocks, respectively, followed by a patch merging layer for downsampling. Inside the SwinT block, window attention and shifted window attention mechanisms are alternately employed, as shown in the Fig. 2. Here, LayerNorm refers to the normalization layer, W-MSA and SW-MSA represent window-based attention and shifted window attention, respectively, and MLP denotes the multi-layer perceptron.

For input images of size $224 \times 224 \times 3$, a convolutional layer with a kernel size of 4×4, stride 4, and padding 0 is applied, resulting in output dimensions of $56 \times 56 \times 96$. After passing through Stage 1, the image size is reduced to $28 \times 28 \times 192$, to $14 \times 14 \times 384$ in Stage 2, and to $7 \times 7 \times 768$ in Stage 3. Stage 4 does not include a patch merging layer, so the image size remains unchanged in this stage. Three cross-attention modules are inserted between the stages, which do not alter the dimensions of the image features. Finally, an average pooling layer reduces the output size to $1 \times 1 \times 768$. The features from the top and bottom streams are concatenated into a 1×1536 vector. The feature vector changes to 1×512 through a fully connected layer with the addition of the BatchNorm layer, Dropout layer, and GELU activation function. The process is repeated to produce the final 1×19 feature vector. BatchNorm helps maintain the inputs of each layer within a relatively stable range, mitigating the gradient vanishing problem. Dropout reduces the interaction between neurons, forcing the network to learn more robust and generalized feature representations, thereby reducing overfitting. GELU is a nonlinear function that introduces non-linearity into the network. Its expression is as follows:

$$GELU\,(x) = x * \int_{-\infty}^{x} \frac{e^{-t^2/2}}{\sqrt{2\pi}} dt \tag{3}$$

Compared to ReLU, the GELU function provides a non-zero gradient near the origin, which can help mitigate gradient vanishing during training.

3.3 BFCA module

To ensure thorough interaction and integration of features from the front and back images of chrysanthemums, this study incorporates a BFCA module between each stage of the SwinT, as illustrated in the Fig. 3.

In this module, the residuals and outputs from the front stream, as well as the residuals from the back stream, are used as the Query (Q), Key (K), and Value (V) inputs for the attention mechanism. The correlation between the front stream's residuals and outputs is computed to obtain attention scores, which are then multiplied by the back stream's residuals to produce the overlapping features between the front and back streams. At this stage, the feature from the front stream is influenced by the feature information from the back image, indicating that the two types of feature information have been integrated. The specific mechanism calculation is as follows:

$$Q_n = X_{res} W^Q \tag{4}$$

$$K_n = X W^K \tag{5}$$

$$V_n = Y_{res} W^V \tag{6}$$

$$CrossAttention\,(Q_n, K_n, V_n) = softmax\left(\frac{Q_n K_n^T}{\sqrt{d_k}}\right) V_n \tag{7}$$

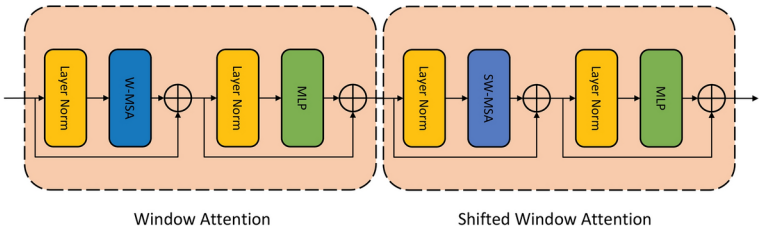


Fig. 2 W-MSA and SW-MSA

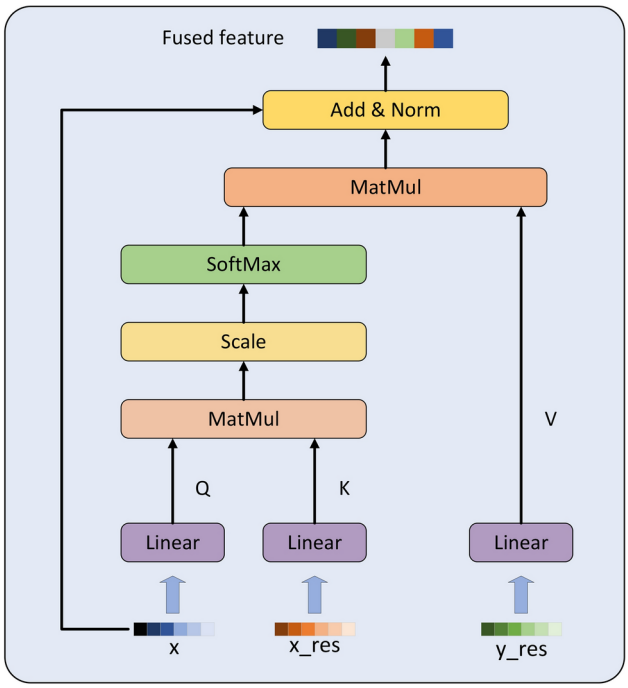


Fig. 3 BFCA module

where X_{res} is the residual of the current stream, X is the output of the previous layer of the current stream, Y_{res} is the residual of the other stream, and W^Q, W^K, and W^V are linear mappings. d_k represents the dimension of d_k . The softmax function is used to normalize the attention scores, converting them into a probability distribution that emphasizes the most relevant features while suppressing less important ones. The final output of the cross-attention layer is:

$$X_{out} = CrossAttention + X_{in} \tag{8}$$

By incorporating skip connections, the original features from the input layer are preserved, allowing the cross-attention layer to focus solely on learning the feature fusion between the two streams. This approach helps mitigate issues related to gradient vanishing and exploding, effectively reducing gradient correlation decay and stabilizing gradients. It also aids in faster model convergence.

This BFCA considers scenarios where different chrysanthemum species might have similar front features but dissimilar back features, or vice versa. This integration of front and back features allows for more accurate species identification.

3.4 Network training

In the model training process, the model is initially trained using the standard Adam optimizer and a cross-entropy loss function. The dataset is divided into training, validation, and test sets in a 6:2:2 ratio. After each iteration, the model is validated on the validation set, and the best-performing model is saved. To reduce experimental errors introduced by random sampling, the entire training and testing process is repeated 50 times, and the average accuracy is used as the final result.

To further improve the model's performance, especially in fine-grained classification tasks, a secondary training strategy is introduced. After every 8 batches of training, we record the model's predicted probability vector $p = [p_1, p_2, \ldots, p_C]$, where C is the number of classes, and p_i represents the probability that a sample belongs to class i for each sample's predicted probability vector p and its corresponding one-hot encoded vector $y = [y_1, y_2, \ldots, y_C]$, we calculate the Euclidean distance d between them using the following formula:

$$d = \sqrt{\sum_{i=1}^{C} (p_i - y_i)^2} \tag{9}$$

The Euclidean distance reflects the deviation between the model's predicted confidence and the actual label. A larger distance indicates that the model struggles more to correctly classify the sample. After calculating the Euclidean distances for all samples, the 24 samples with the largest distances are selected as difficult-to-classify samples. These samples are then reassembled into a new batch for an additional round of training. This secondary training method aims to strengthen the model's ability to learn from challenging samples, thereby enhancing its capacity to differentiate between similar categories. The pseudocode for this process is as follows:

Initialize model and training parameters
Initialize an empty list `Distance_List` to store Euclidean distances for each sample
Set `num_batches` to the length of batches awaiting processing
for $batch_index \leftarrow 1$ **to** `num_batches` **do**
Get the current batch corresponding to `batch_index`
Perform forward propagation
Compute loss
Perform backward propagation
Update model weights
Initialize an empty list `Batch_Distances` for current batch
  for $sample_index \leftarrow 1$ **to** 24 **do**
Get the current sample corresponding to `si` ▷ $sample_index$, abbreviated as si
Compute predicted probability vector $\mathbf{p}^{(si)} = [p_1, p_2, \ldots, p_C]$
Compute one-hot encoded vector $\mathbf{y}^{(si)} = [y_1, y_2, \ldots, y_C]$
Compute Euclidean distance $\mathbf{d}^{(si)} = \sqrt{\sum_{i=1}^{C}(p_i - y_i)^2}$
Append (current sample, $\mathbf{d}$) to `Batch_Distances`
  end for
Append `Batch_Distances` to `Distance_List`
  if $batch_index\%8 == 0$ **then**
Flatten `Distance_List`
Sort `Distance_List` by $\mathbf{d}$ in **descending** order
Select top 24 samples
Create new batch from top samples
Perform secondary training with the new batch
Clear `Distance_List`
  end if
end for

Algorithm 1 Secondary training strategy

4 Experiment

4.1 Dataset and evaluation metric

The dataset for this experiment includes chrysanthemum images from various regions and categories, featuring 18 types from 17 counties across 8 provinces, including Anhui, Henan, and Jiangsu. Each type has 150 images of both the front and back, amounting to approximately 5700 images in total. The images are paired as front and back views, forming 2765 samples. These are divided into training, validation, and test sets in a 6:2:2 ratio. Figure 4 illustrates sample images of chrysanthemums from different regions and categories.

This study evaluates the performance of chrysanthemum image classification methods using five metrics: Accuracy, Recall, Precision, F1 Score, and Standard Deviation (STD) across classes. Here, T and F indicate whether the predicted category matches the actual category, and P and N specify whether the prediction is positive or negative. Accordingly, TP represents true positives, FP represents false positives, FN represents false negatives, and TN represents true negatives.

Accuracy is the ratio of correctly predicted samples to the total number of samples. It is calculated using the formula:

$$ACC = \frac{TP + TN}{P + N} \tag{10}$$

Precision is the ratio of correctly predicted positive samples to the total number of predicted positive samples. It is calculated using the formula:

$$precision = \frac{TP}{TP + FP} \tag{11}$$

Recall is the ratio of correctly predicted positive samples to the total number of actual positive samples. It is calculated using the formula:

$$Recall = \frac{TP}{TP + FN} \tag{12}$$

The F1 Score assesses the accuracy of a binary classification model and is the harmonic mean of Precision and Recall, ranging from 0 to 1. It is calculated using the formula:

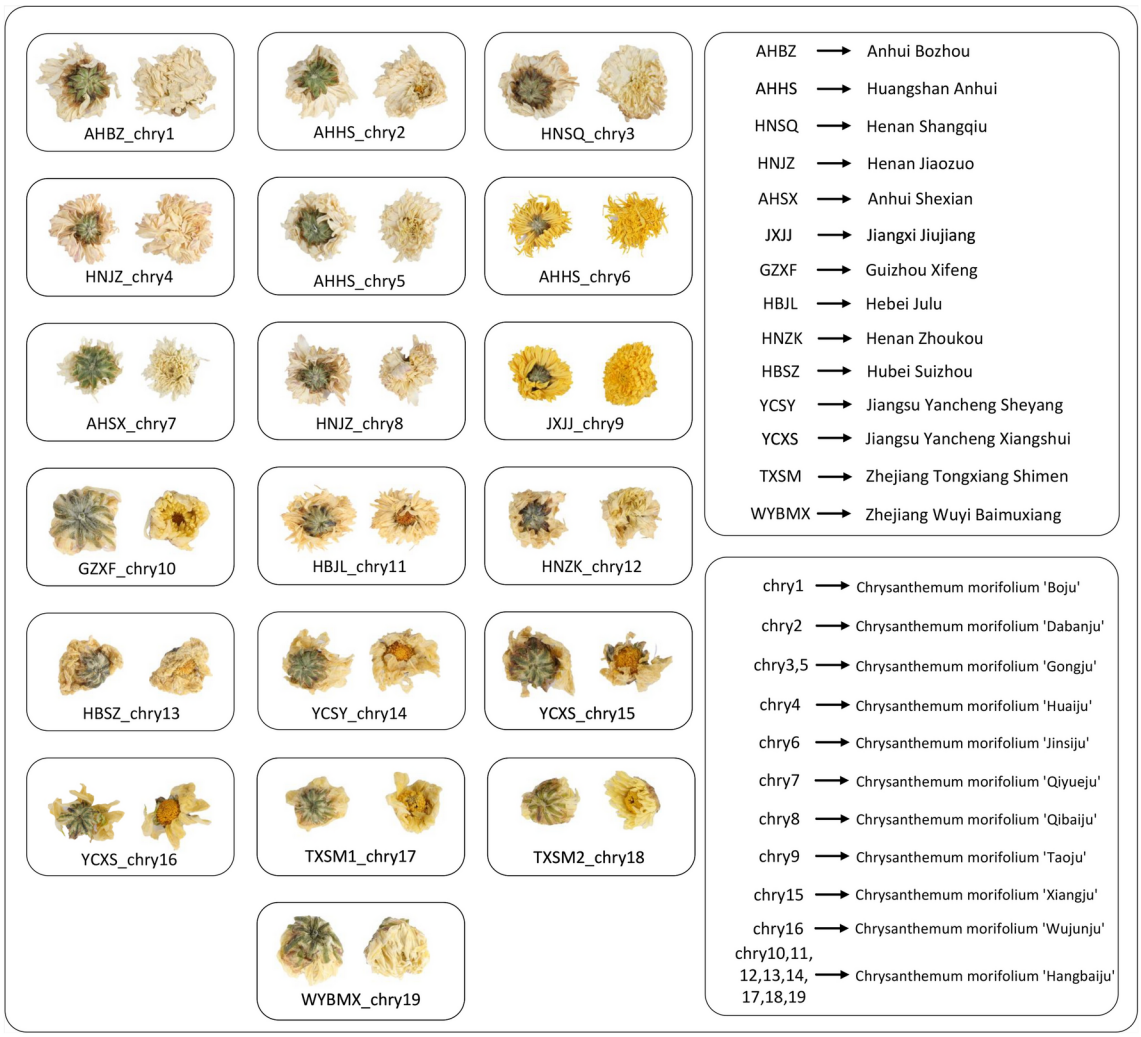


Fig. 4 Chrysanthemum illustration

$$F1 = \frac{2 * TP}{2 * TP + FN + FP} = \frac{2 * precision * Recall}{Precision + Recall} \tag{13}$$

Since chrysanthemum image classification is a multi-class problem, the metrics in this study aggregate the metrics for each class. Macro metrics are the averages of the evaluation metrics across all classes, enabling the calculation of macro Precision, macro Recall, and macro F1 Score. Their formulas are as follows:

$$Pr_{macro} = \frac{1}{n}\sum_{i=1}^{n} Pr_i \tag{14}$$

$$R_{macro} = \frac{1}{n}\sum_{i=1}^{n} R_i \tag{15}$$

$$F_{1_{marco}} = 2 \cdot \frac{Pr_{marco} \times R_{marco}}{Pr_{marco} + R_{marco}} \tag{16}$$

Higher values of these metrics signify better classification performance. The Standard Deviation between classes reflects the method's variance and balance across different chrysanthemum types. A lower STD indicates more stable classification performance. The formula for STD is:

$$STD = \sqrt{\frac{\sum_{i=1}^{n} (Pr_i - Pr_{macro})^2}{n}} \tag{17}$$

Comparing the measurement results of different models provides a comprehensive understanding of the classifier's performance in chrysanthemum image classification tasks. Notably, since our method leverages both the front and back images of chrysanthemums for training, we also applied a two-stream mirror setup in the comparison experiments for other methods. Specifically, we designed a two-stream network architecture for each comparison method, where the upper and lower paths receive the front and back

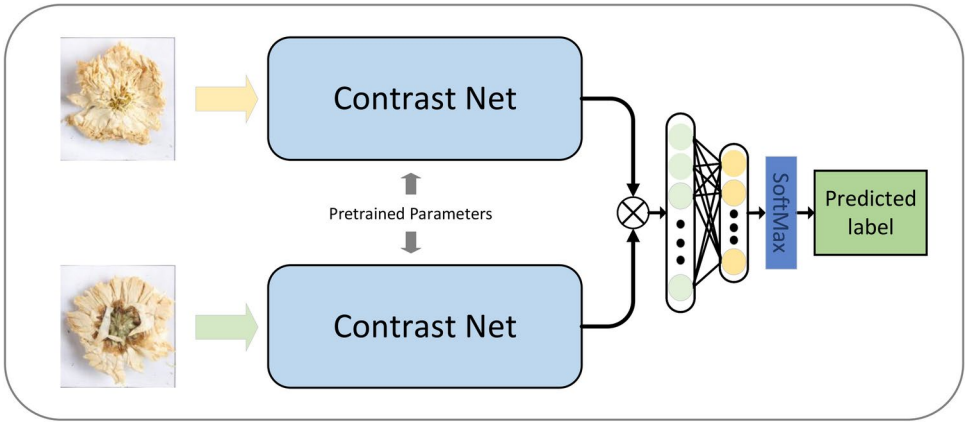


Fig. 5 Network architecture of the comparison methods

Table 1 Performance comparisons of different image classification methods

BackBone	Method	Acc(%)	Recall(%)	Precision(%)	F1(%)	STD(%)
Traditional networks	EfficientNetV2	83.82	84.27	84.08	84.18	12.59
	RegNet	87.44	87.80	87.72	87.76	11.88
	ConvNeXtV2	90.85	91.10	91.06	91.08	10.33
ViTs	EfficientViT	83.02	83.49	83.42	83.46	12.69
	RepViT	88.91	89.20	89.18	89.19	10.98
	FastViT	88.44	88.73	88.71	88.72	10.30
	Deit3	87.70	88.03	87.98	88.01	10.70
	Ours	93.95	94.11	94.15	94.13	7.18

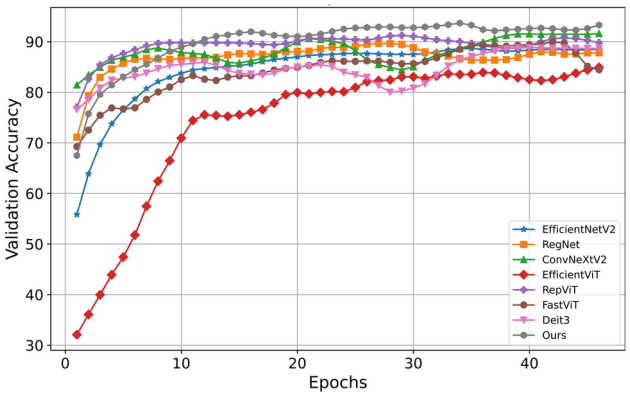


Fig. 6 Convergence speeds of different methods

images, respectively. All models were evaluated using pretrained parameters from ImageNet to ensure a fair comparison regarding resource usage and model complexity. This approach allows the model to capture information from both sides of the chrysanthemum and enhance classification performance by integrating information from both paths. The two-stream network structure of the compared models is illustrated in Fig. 5.

4.2 Performance comparision

To validate the effectiveness of the proposed method, we compared it with seven mainstream image classification techniques. These include three traditional convolutional neural network models: EfficientNetV2, RegNet, and ConvNeXtV2, as well as four ViT models: Deit3, FastViT, RepViT, and EfficientViT. Given that we used the smallest version of SwinT as the backbone, we also selected the corresponding smaller versions of other models for comparison, such as EfficientNetV2-s.

The experimental results are presented in Table 1. The proposed method demonstrates superior performance across all evaluation metrics, with Recall, Precision, and F1 Score each exceeding 94%, outperforming the second-best model, ConvNeXtV2, by 3%. Furthermore, our model shows outstanding performance in the standard deviation between classes, at just 7.18%, which is 3% to 5% lower than that of other models. This highlights that our model excels not only in overall performance but also in stability and balance. While other models such as RepViT, FastViT, and Deit3 have comparable performance, they do not match the overall effectiveness of our custom model. In conclusion, the custom model outperforms others in both overall performance and stability.

On the other hand, we compared the iterative performance of different image classification models during the training process. The results are depicted in Fig. 6, where the x-axis represents the number of iterations and the y-axis indicates accuracy on the validation set. Based on Fig. 6, our model outperforms the other methods. Although RegNet and RepViT converge more quickly in the early stages of training, our proposed method catches up and surpasses

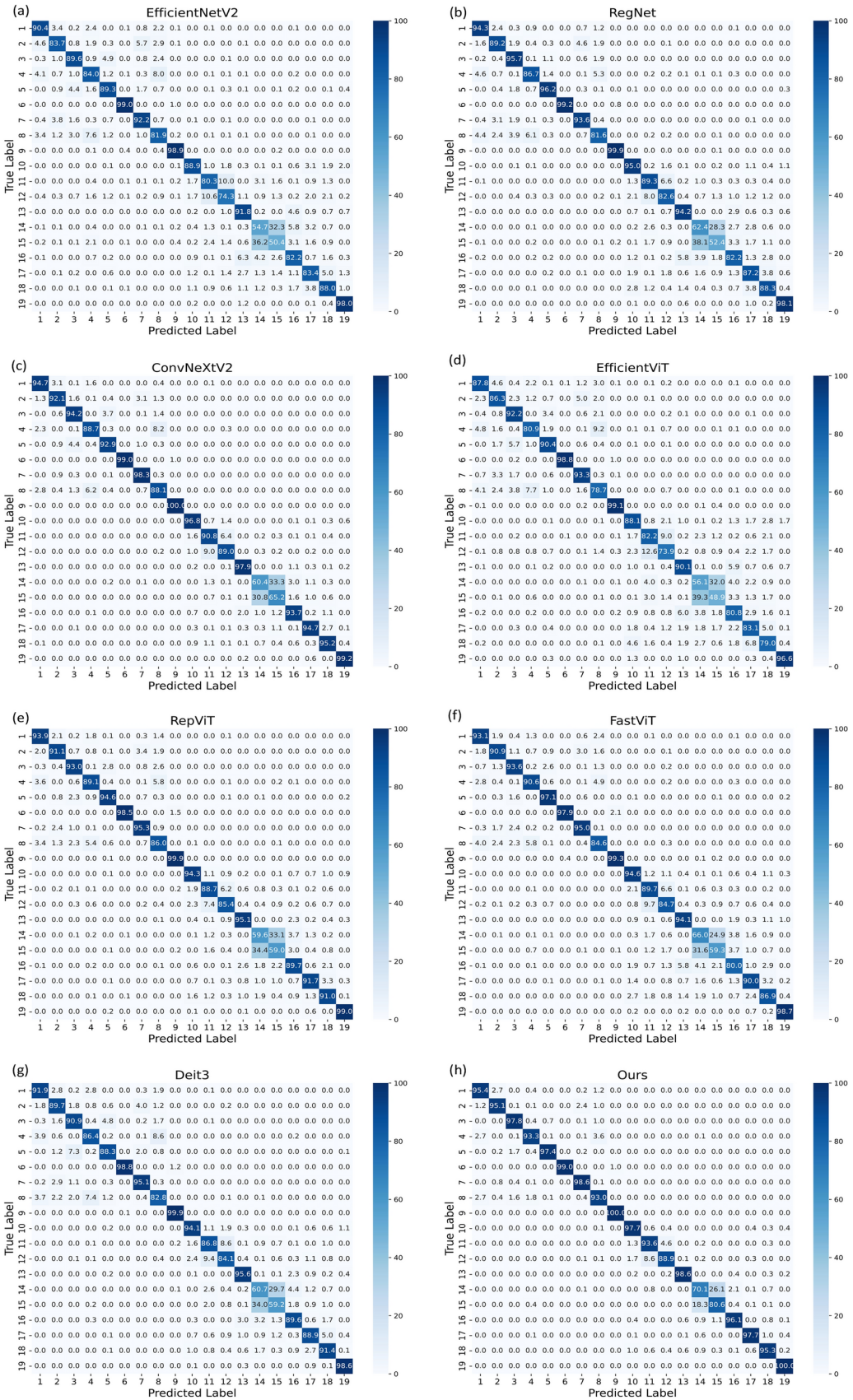


Fig. 7 Confusion matrix of different methods

them around the 10th epoch. Although ConvNeXtV2's final results are similar to those of our model, its accuracy curve shows more fluctuations. In contrast, our model exhibits smoother and more reliable performance throughout the training process. These findings highlight the advantages of our method in terms of model performance and training efficiency, demonstrating rapid accuracy gains in the early stages and maintaining high stability in the later stages, ultimately achieving excellent classification performance.

4.3 Performance on individual classification

To more clearly illustrate the accuracy of different methods across various categories, we have plotted the confusion matrix shown in the Fig. 7. From this chart, it is apparent that the proposed method achieves over 90% accuracy in most categories. However, the accuracy is somewhat lower for categories chry14 and chry15, at 70% and 80%, respectively. In comparison, other methods achieve only 50% to 60% accuracy in these categories and are prone to mutual misclassification, making it nearly impossible to distinguish between these two similar categories from Yancheng Sheyang. In other categories, the proposed method shows strong competitive performance. Although its accuracy in category chry12 is slightly below ConvNeXtV2's 89.0%, it achieves the highest accuracy in all other categories. In the 19×19 matrix, the proposed method has a misclassification rate of 0% in 254 cells, outperforming 212 cells of Deit3 and 237 cells of ConvNeXtV2. This demonstrates the proposed method's significant advantage in reducing misclassifications. Thus, the proposed method excels not only in most categories but also shows considerable strength in handling complex categories. Overall, while other models perform weaker in some categories, the proposed method exhibits exceptional performance in chrysanthemum classification tasks, especially in distinguishing highly similar categories.

4.4 Stability analysis

To assess the stability of the proposed method, we created box plots comparing its performance with other methods in chrysanthemum image classification. In these plots, the height of the box represents the interquartile range (IQR), which shows the spread of the central 50% of the data. The red line within the box indicates the median value. The whiskers extending from the box capture the range of the data

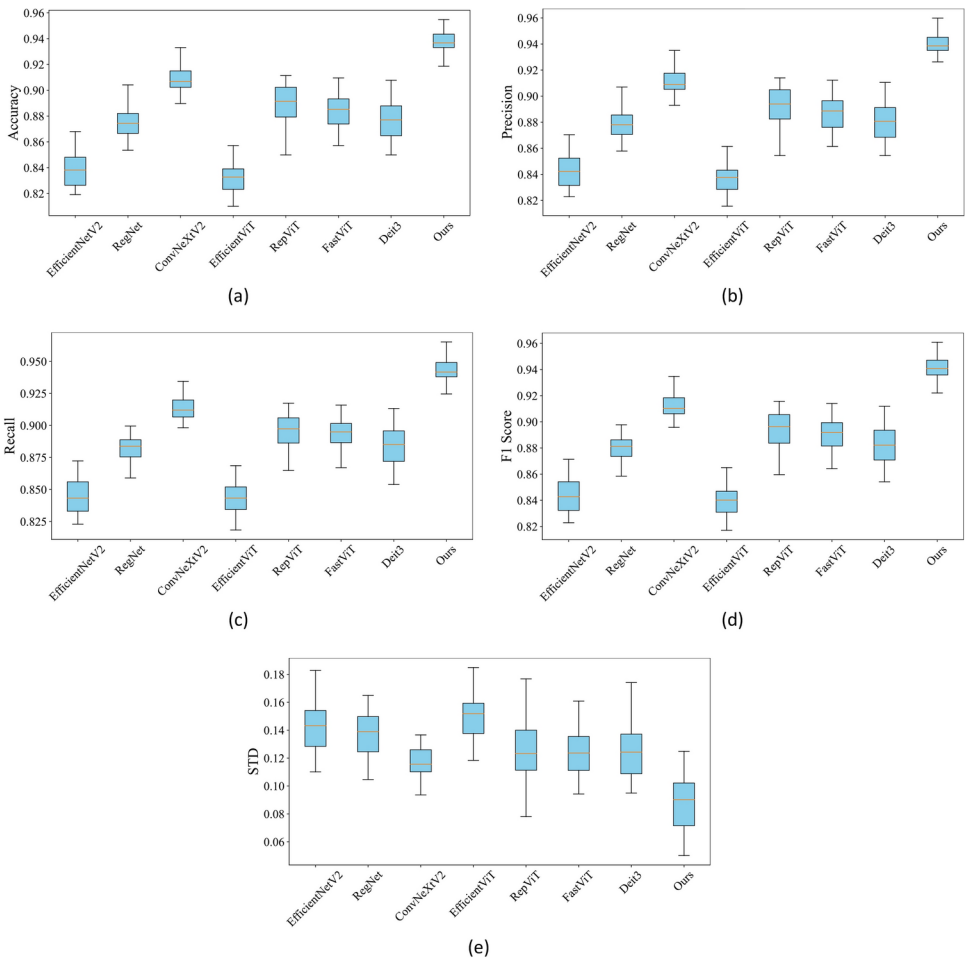


Fig. 8 Boxplots displaying various metrics for different methods in chrysanthemum image classification

within 1.5 times the IQR from the quartiles. As illustrated in Fig. 8, the x-axis represents the different image classification methods, and the y-axis indicates the evaluation metrics for each method. The shape of the boxes reflects the data distribution for each method's results.

Based on the results shown in figures (a–d), the proposed method demonstrates superior stability across these four metrics compared to other methods. Although ConvNeXtV2 performs relatively well in terms of stability, its overall performance on various metrics does not match that of the proposed method. Figure (e) reveals that the proposed method's stability regarding the standard deviation between categories is slightly lower than that of other methods, as indicated by the larger box plot. In contrast, other methods show smaller boxes, but their standard deviation values are still higher than those of the proposed method, which has a better ability to distinguish between certain categories. Overall, the proposed method combines high stability and accuracy, proving effective for chrysanthemum image classification.

4.5 Robustness analysis

To simulate potential distortion issues present in real production environments, we performed a series of distortion simulations on the original chrysanthemum images. These distortions include, but are not limited to, Gaussian blur, motion blur, and brighten. During the experiment, we first trained the model on the normal dataset. We then applied the trained model to both the original images and those subjected to the various distortions mentioned. By extracting the feature vectors produced by the model and applying SoftMax processing, we obtained the predicted probabilities for different categories. This approach aims to assess the model's robustness and adaptability to different types of distortions. The experimental results are illustrated in Fig. 9.

As illustrated in the figure, we tested the original and distorted images of three chrysanthemum flowers and compared the performance of ConvNeXtV2, RepViT, and FastViT. The analysis focuses on the probability of each model correctly identifying the categories. The red values indicate the degree of performance degradation under different distortions, while the brown values show the difference from the highest probability after excluding the current category. Check marks in the figure indicate that the model output the highest probability, thus correctly identifying the chrysanthemum category, while crosses denote cases where the highest probability did not correspond to the correct category, resulting in an incorrect prediction.

The results show that all models experience varying degrees of performance decline after distortion simulation. Firstly, FastViT exhibits some resistance to motion blur, with only an average 25% performance drop, but struggles significantly with Gaussian blur and brighten, with performance degradation surpassing 76.72% and 87.24% respectively, leading to complete recognition failure. Secondly, RepViT shows a similar pattern, with some resistance to motion blur and Gaussian blur in chry14, where the performance drops by approximately 20.83% and 20.72%, but performance drops sharply by over 40% for other chrysanthemum categories with distortions. Thirdly, ConvNeXtV2 performs well with motion blur and brighten, with

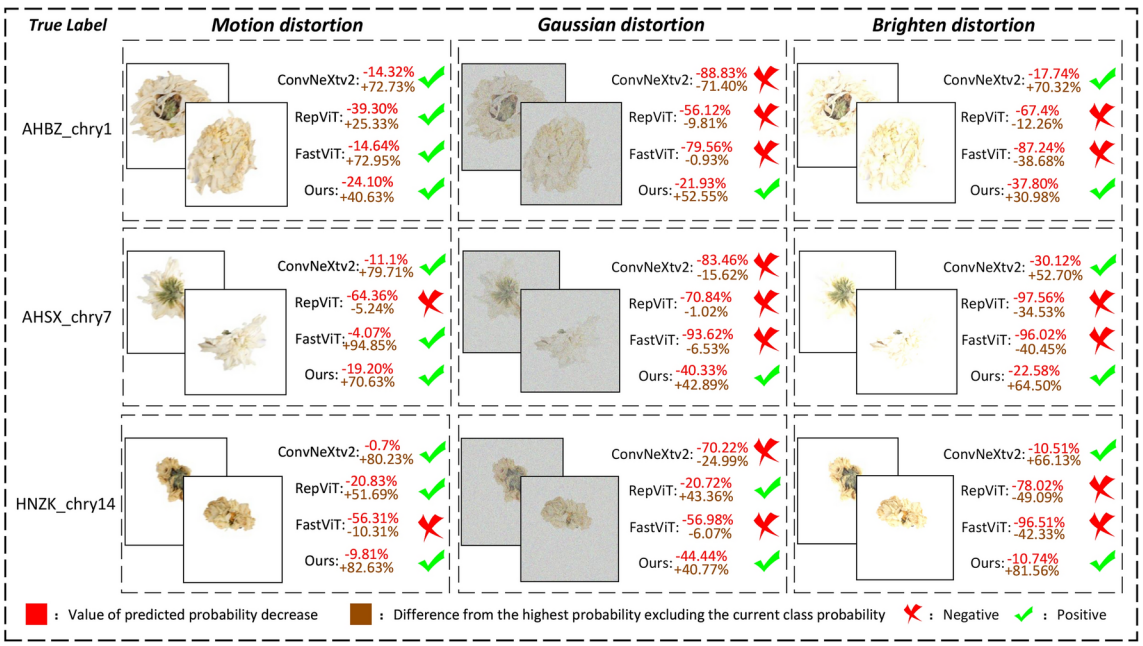


Fig. 9 Test results for different distorted chrysanthemum images

Table 2 Experimental results of ablation study of two stream

Method	Acc(%)	Recall(%)	Precision(%)	F1(%)
Front image only	85.14	85.55	85.90	85.72
Back image only	88.66	88.94	89.05	89.00
Both front and back image	93.95	94.11	94.15	94.13

Table 3 Experimental results of ablation study of secondary training

Secondary trained	Acc(%)	Recall(%)	Precision(%)	F1(%)
✖	93.29	93.45	93.47	93.46
✓	93.95	94.11	94.15	94.13

performance declines kept within 14.32% and 30.12%, but shows inadequate results with Gaussian blur, experiencing a notable 70.22% drop in predicted probabilities. Finally, the proposed method exhibits the smallest performance drop across various distortions, with declines never exceeding 44.44%. It accurately identifies chrysanthemum types, maintaining a considerable margin over the second-highest predicted probability, often surpassing 30%, which demonstrates superior recognition capability and robustness. These results underscore the proposed model's exceptional adaptability and robustness in handling various distortions, surpassing existing methods.

4.6 Ablation study

To assess the contribution of the dual-front-and-back approach in the proposed chrysanthemum classification method, we conducted a series of ablation experiments. These experiments included: (1) using only front images of chrysanthemums, (2) using only back images of chrysanthemums, and (3) using both front and back images. In these single-path experiments, the network utilized the small version of the SwinT, with results detailed in Table 2.

From the results presented in the table, we can conclude the following: Using only front images of chrysanthemums yields poor classification results, with an accuracy of just 85.14%. This indicates that front image features are relatively limited and cannot fully capture all characteristics of the chrysanthemum, resulting in subpar classification performance. In contrast, using only back images improves performance, with accuracy increasing to 88.66%. This suggests that back image features are richer and more effective at enhancing classification performance, although they still do not fully leverage the model's potential. When both front and back images are used, classification performance improves significantly, achieving an accuracy of 93.95%. This demonstrates that combining features from both sides complements each other, greatly enhancing the model's performance. In summary, both front and back image features offer distinct advantages, and their combined use provides a more comprehensive representation of chrysanthemum characteristics, leading to superior classification performance.

Furthermore, to validate the effectiveness of the proposed secondary training method, we conducted a series of ablation experiments. We tested the method both with and without the secondary training approach, with results shown in Table 3.

The results show that the secondary training method led to improvements across all performance metrics. Accuracy increased from 93.29% to 93.95%, recall from 93.45% to 94.11%, precision from 93.47% to 94.15%, and F1 score from 93.46% to 94.13%. Furthermore, the standard deviation

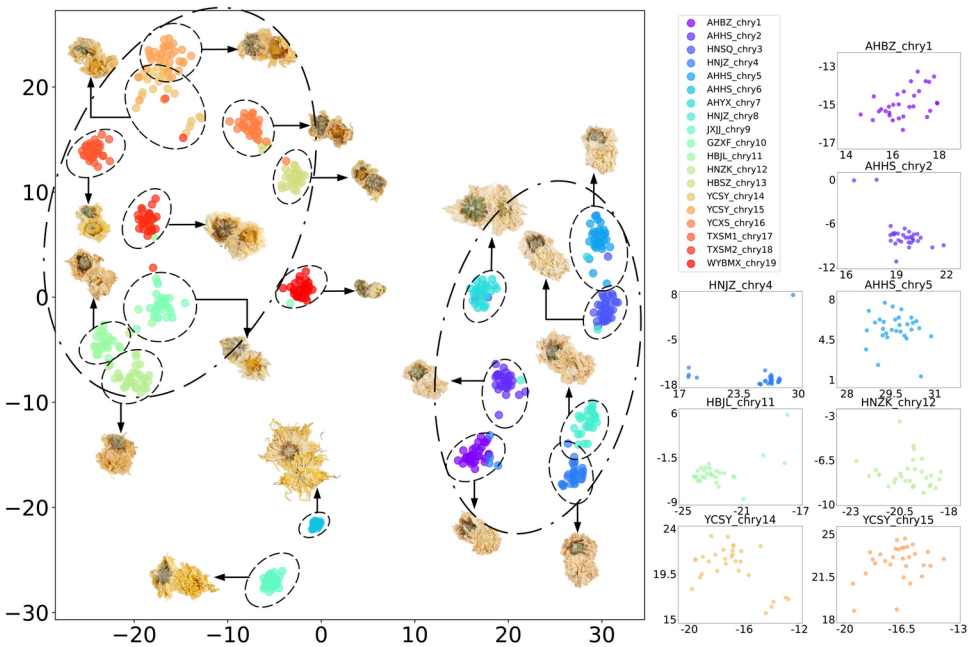


Fig. 10 Visual clustering scatter plot for chrysanthemum classification

decreased from 7.75% to 7.18%, reflecting improved model stability. These results validate the effectiveness of the secondary training method in enhancing model generalization and significantly improving its ability to recognize challenging categories.

4.7 Visual analysis

This paper utilizes t-SNE to generate a clustering scatter plot for chrysanthemum classification, depicting the distribution of samples in the dataset, as shown in Fig. 10. The x-axis and y-axis represent two specific feature attributes, with each scatter point representing an individual chrysanthemum sample. The subplot on the right displays scatter plots for several similar categories. Different colors indicate different chrysanthemum categories, revealing their distribution in the feature space. The plot shows two main clusters of chrysanthemums, with several smaller sub-clusters within these larger clusters, displaying similar shapes and corresponding to different chrysanthemum categories. It is evident from the plot that the scatter points for YCSY_chry14 and YCSY_chry15 overlap, suggesting that these similar chrysanthemum types have very similar features, making them difficult to differentiate. Other categories, such as AHBZ_chry1,HNJZ_chry4, and AHHS_chry5, exhibit similar overlapping patterns. This visualization provides valuable insights into the differences between chrysanthemum categories and aids in optimizing classification models to better distinguish these categories.

5 Conclusion

This paper proposes a chrysanthemum classification method based on a two-stream Swin Transformer network and a BFCA module. By preprocessing front and back images of chrysanthemums, the method effectively integrates features using a cross-attention mechanism and residual modules. These designs, along with a secondary training strategy, enhance the model's generalization performance and ensure thorough feature fusion. Specially, the proposed method achieves outstanding performance, with accuracy, precision, recall, and F1 scores exceeding 93.95%, surpassing other models by at least 3%. The standard deviation result of 7.18, along with the box plot results, demonstrates the stability of the method. The method also keeps stable under various distortions. Meanwhile, our method converged in approximately 10 epochs, which is faster than other methods. In summary, the method excels in accuracy, stability, robustness, and convergence speed, providing an effective solution for chrysanthemum classification. In future research, we will concentrate on deep fusion strategies combining CNN and ViT to harness the strengths of CNN in local feature extraction and the capabilities of ViT in global information capture. This approach aims to further improve the model's ability to perceive detailed features of both the front and back of chrysanthemums.

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Author contributions Conceived and designed the experiments: Xichen Yang and Yifan Chen. Performed the experiments: Xichen Yang and Yifan Chen. Analyzed the data: Hui Yan, Yifan Chen, Jia Liu, Jian Jiang, Zhongyuan Mao. Wrote and reviewed the paper: Xichen yang, Hui Yan, Yifan Chen, Tianshu Wang. Approved the final version of the paper: Xichen yang, Hui Yan, Yifan Chen, Tianshu Wang.

Data Availability Data supporting the results of this study are available from the corresponding author upon reasonable request, including source code and experimental results.

Declarations

Conflict of interest The authors have declared that there are no Conflict of interest exist.

Ethical approval Not applicable.

Research involving human and/or animal rights The authors have no Conflict of interest to disclose. Further, the authors certify that, the research presented in this article does not involve any human participants or animals. Moreover, all the datasets used in our experiment are public.

Consent to participate Not applicable.

Consent for publication Not applicable.

References

1. Sun J, Wang Z, Chen L, Sun G (2021) Hypolipidemic effects and preliminary mechanism of chrysanthemum flavonoids, its main components luteolin and luteoloside in hyperlipidemia rats. Antioxidants 10(8):1309
2. Zhan J, He F, Cai H, Wu M, Xiao Y, Xiang F, Yang Y, Ye C, Wang S, Li S (2021) Composition and antifungal mechanism of essential oil from Chrysanthemum morifolium cv. fubaiju. J Funct Foods 87:104746
3. Jiang S, Wang M, Jiang Z, Zafar S, Xie Q, Yang Y, Liu Y, Yuan H, Jian Y, Wang W (2021) Chemistry and pharmacological activity of sesquiterpenoids from the chrysanthemum genus. Molecules 26(10):3038
4. Youssef FS, Eid SY, Alshammari E, Ashour ML, Wink M, El-Readi MZ (2020) Chrysanthemum indicum and chrysanthemum morifolium: chemical composition of their essential oils and their

potential use as natural preservatives with antimicrobial and antioxidant activities. Foods 9(10):1460
5. He J, Zhu S, Chu B, Bai X, Xiao Q, Zhang C, Gong J (2019) Nondestructive determination and visualization of quality attributes in fresh and dry chrysanthemum morifolium using near-infrared hyperspectral imaging. Appl Sci 9(9):1959
6. Chen Y, Zhen X-T, Yu Y-L, Shi M-Z, Cao J, Zheng H, Ye L-H (2021) Chemoinformatics based comprehensive two-dimensional liquid chromatography-quadrupole time-of-flight mass spectrometry approach to chemically distinguish chrysanthemum species. Microchem J 168:106464
7. Chen Y, Zhen X-T, Yu Y-L, Shi M-Z, Cao J, Zheng H, Ye L-H (2021) Chemoinformatics based comprehensive two-dimensional liquid chromatography-quadrupole time-of-flight mass spectrometry approach to chemically distinguish chrysanthemum species. Microchem J 168:106464
8. Zhai G, Li Z, Lu W, Zhao Y, Wang C (2016) Study on varieties of identification ornamental chrysanthemum based on image processing. J Chin Agric Mech 37(2):105–110
9. Long W, zhang Q, Wang S-R, Suo Y, Chen H, Bai X, Yang X, Zhou Y-P, Yang J, Fu H (2023) Fast and non-destructive discriminating the geographical origin of Hangbaiju by hyperspectral imaging combined with chemometrics. Spectrochim Acta Part A: Mol Biomol Spectrosc 284:121786
10. Liu Z, Gao K, Tian y, Dai S, Song X (2017) Identification of Chrysanthemum cultivars based on unfolding image with lbp texture feature. Research progress of ornamental horticulture in China
11. Hadipour-Rokni R, Asli-Ardeh EA, Jahanbakhshi A, Sabzi S et al (2023) Intelligent detection of citrus fruit pests using machine vision system and convolutional neural network through transfer learning technique. Comput Biol Med 155:106611
12. Momeny M, Jahanbakhshi A, Neshat AA, Hadipour-Rokni R, Zhang Y-D, Ampatzidis Y (2022) Detection of citrus black spot disease and ripeness level in orange fruit using learning-to-augment incorporated deep networks. Eco Inform 71:101829
13. Azadnia R, Fouladi S, Jahanbakhshi A (2023) Intelligent detection and waste control of hawthorn fruit based on ripening level using machine vision system and deep learning techniques. Res Eng 17:100891
14. Momeny M, Neshat AA, Jahanbakhshi A, Mahmoudi M, Ampatzidis Y, Radeva P (2023) Grading and fraud detection of saffron via learning-to-augment incorporated inception-v4 cnn. Food Control 147:109554
15. Zhang Z, Xiao J, Wang W, Zielinska M, Wang S, Liu Z, Zheng Z (2024) Automated grading of angelica sinensis using computer vision and machine learning techniques. Agriculture 14(3):507
16. Zhang Z, Xiao J, Wang S, Wu M, Wang W, Liu Z, Zheng Z (2023) Origin intelligent identification of angelica sinensis using machine vision and deep learning. Agriculture 13(9):1744
17. Xie Z, Zhang W, Sheng B, Li P, Chen CP (2021) Bagfn: broad attentive graph fusion network for high-order feature interactions. IEEE Trans Neural Netw Learn Syst 34(8):4499–4513
18. Ali SG, Wang X, Li P, Li H, Yang P, Jung Y, Qin J, Kim J, Sheng B (2024) Egdnet: an efficient glomerular detection network for multiple anomalous pathological feature in glomerulonephritis. Vis Comput. https://doi.org/10.1007/s00371-024-03570-5
19. Krizhevsky A, Sutskever I, Hinton GE (2012) Imagenet classification with deep convolutional neural networks. Adv Neural Inform Process Syst 25:1
20. Simonyan K, Zisserman A (2014) Very deep convolutional networks for large-scale image recognition. arXiv preprint arXiv:1409.1556
21. Szegedy C, Liu W, Jia Y, Sermanet P, Reed S, Anguelov D, Erhan D, Vanhoucke V, Rabinovich A (2015) Going deeper with convolutions. In: Proceedings of the IEEE conference on computer vision and pattern recognition, pp 1–9
22. He K, Zhang X, Ren S, Sun J (2016) Deep residual learning for image recognition. In: Proceedings of the IEEE conference on computer vision and pattern recognition, pp 770–778
23. Huang G, Liu Z, Van Der Maaten L, Weinberger KQ (2017) Densely connected convolutional networks. In: Proceedings of the IEEE conference on computer vision and pattern recognition, pp 4700–4708
24. Xu J, Pan Y, Pan X, Hoi S, Yi Z, Xu Z (2022) Regnet: Self-regulated network for image classification. IEEE Trans Neural Netw Learn Syst 34(11):9562–9567
25. Wang C-Y, Liao H-YM, Wu Y-H, Chen P-Y, Hsieh J-W, Yeh I-H (2020) Cspnet: A new backbone that can enhance learning capability of cnn. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition workshops, pp 390–391
26. Tan M (2019) Efficientnet: Rethinking model scaling for convolutional neural networks. arXiv preprint arXiv:1905.11946
27. Woo S, Debnath S, Hu R, Chen X, Liu Z, Kweon IS, Xie S (2023) Convnext v2: Co-designing and scaling convnets with masked autoencoders. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 16133–16142
28. Liu Z, Mao H, Wu C-Y, Feichtenhofer C, Darrell T, Xie S (2022) A convnet for the 2020s. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 11976–11986
29. He K, Chen X, Xie S, Li Y, Dollár P, Girshick R (2022) Masked autoencoders are scalable vision learners. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 16000–16009
30. Yuan P, Qian S, Zhai Z, FernánMartínez J, Xu H (2022) Study of chrysanthemum image phenotype on-line classification based on transfer learning and bilinear convolutional neural network. Comput Electron Agric 194:106679
31. Yuan P, Ding Y, Xu H (2024) Fine-grained chrysanthemum phenotype recognition based on deep active learning and cbam. Trans Chin Soc Agric Mach 55(2):258–267
32. Liu Z, Wang J, Tian Y, Dai S (2019) Deep learning for image-based large-flowered chrysanthemum cultivar recognition. Plant Methods 15:1–11
33. Wang J, Tian Y, Zhang R, Liu Z, Tian Y, Dai S (2022) Multi-information model for large-flowered chrysanthemum cultivar recognition and classification. Front Plant Sci 13:806711
34. Huang S, Liu G (2023) Research on fine-grained classification of chrysanthemum images based on multi-scale and multi-parallel convolutional neural network. In: International conference on cyber security, artificial intelligence, and digital economy (CSAIDE 2023), 12718, pp 430–437. SPIE
35. Zhang R, Tian Y, Zhang J, Dai S, Hou X, Wang J, Guo Q (2021) Metric learning for image-based flower cultivars identification. Plant Methods 17(1):65
36. Hu J, Shen L, Sun G (2018) Squeeze-and-excitation networks. In: Proceedings of the IEEE conference on computer vision and pattern recognition, pp 7132–7141
37. Woo S, Park J, Lee J-Y, Kweon IS (2018) Cbam: Convolutional block attention module. In: Proceedings of the European conference on computer vision (ECCV), pp 3–19
38. Vaswani A (2017) Attention is all you need. Adv Neural Inform Process Syst
39. Dosovitskiy A (2020) An image is worth 16x16 words: transformers for image recognition at scale. arXiv preprint arXiv:2010.11929
40. Touvron H, Cord M, Douze M, Massa F, Sablayrolles A, Jégou H (2021) Training data-efficient image transformers & distillation through attention. In: International conference on machine learning, pp 10347–10357. PMLR

41. Wu K, Zhang J, Peng H, Liu M, Xiao B, Fu J, Yuan L (2022) Tinyvit: Fast pretraining distillation for small vision transformers. In: European conference on computer vision. Springer, pp 68–85
42. Liu Z, Lin Y, Cao Y, Hu H, Wei Y, Zhang Z, Lin S, Guo B (2021) Swin transformer: Hierarchical vision transformer using shifted windows. In: Proceedings of the IEEE/CVF international conference on computer vision, pp 10012–10022
43. Vasu PKA, Gabriel J, Zhu J, Tuzel O, Ranjan A (2023) Fastvit: A fast hybrid vision transformer using structural reparameterization. In: Proceedings of the IEEE/CVF international conference on computer vision, pp 5785–5795
44. Liu X, Peng H, Zheng N, Yang Y, Hu H, Yuan Y (2023) Efficientvit: Memory efficient vision transformer with cascaded group attention. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 14420–14430
45. Wang A, Chen H, Lin Z, Han J, Ding G (2024) Repvit: Revisiting mobile cnn from vit perspective. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 15909–15920
46. Lin X, Sun S, Huang W, Sheng B, Li P, Feng DD (2021) Eapt: efficient attention pyramid transformer for image processing. IEEE Trans Multimedia 25:50–61

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